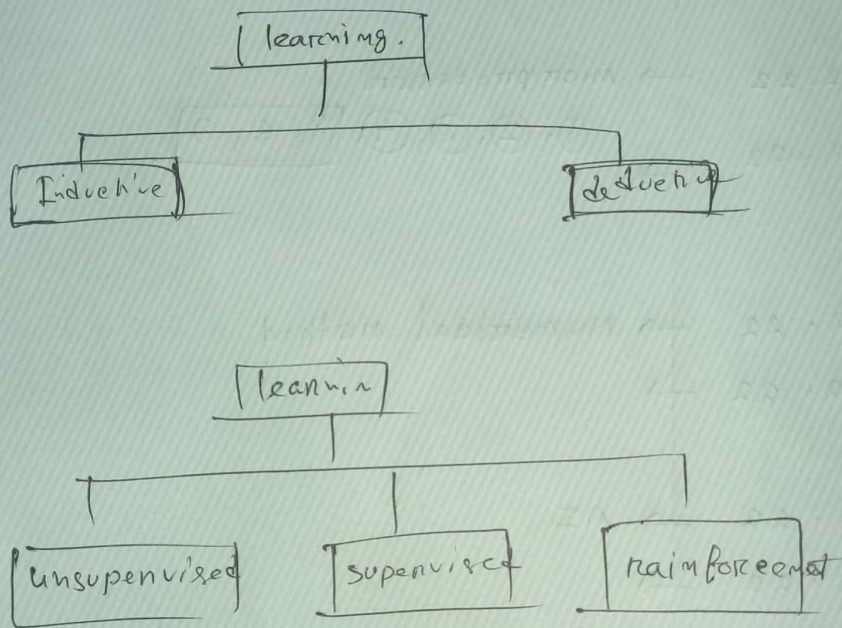




supervised learning

1. training data^{set} (Attribute, feature level information)
2. test data set

generalized.

what is classification and what is regression.

Hypothesis space

How do we choose from among multiple consistent hypotheses?

Learning decision trees (build)

Choosing attribute
entropy, gain
↓
uncertainty

CH = 05

agent :
goal-based-agent :

A problem can be classified formally
by 5 components :

1. Initial state
2. Action
3. Transition
4. Goal test
5. Path costing.

def: Heuristic function

CH-04

✓ def. → local search, local optimization.

✓ Local search for optimization.

eff-cc

constraints → restriction, condition.

→ have an alternate ways,
but bound to do the task
in a certain way.

esp (constraint satisfaction problem)

↳ to solve a wide variety of problems
efficiently.

problem have those things.

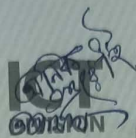
1. variable
2. values
3. constraints



DIGITAL
BANGLADESH
Skilled • Equipped • Digital Ready



FUTURE IS HERE



SKILL DEVELOPMENT FOR
MOBILE
GAME
APPLICATION



মুজিব MUJIB
শতবর্ষ 100

CH-08

First order logic

□ Why propositional logic is the foundation for AI?

□ Difference between propositional logic and first order logic.



□ FOL $\rightarrow$ syntax
 $\rightarrow$ semantics

FOL $\rightarrow$ properties
 $\rightarrow$ Relation
 $\rightarrow$ Function

Combining the best of formal and natural languages

1. Object
2. Relation
3. Function

✓ Quantifiers

$\rightarrow$ universal $\forall$ $(\rightarrow)$
 $\rightarrow$ Existential $\exists$ $(\wedge)$

Sentence to logic / logic to sentence



mobileSkill



gameapp.gov.bd

০৪-১১
Quantifying uncertainty

▣ Probability with a proposition:

▣ Conditional probability:

marginalization
summing out

▣ joint probability
Normalization

▣ Independence probability.

০৮০৭ (Introduction to
Reasoning with uncertainty)

Let

1. uncertainty
2. probability
3. Random variable
- 4.

- ✓ Why uncertainty
- ✓ prior probability
- ✓ Identically independent (II)

$$A \cap B = A \cap B \rightarrow \text{intersection}$$

$$A \cup B = A \cup B \rightarrow \text{union}$$

$$\bar{A} = \neg A \rightarrow \text{complement}$$

$$\Omega = t \rightarrow \text{certain}$$

$$\emptyset = f \rightarrow \text{impossible}$$

1. Joint probability
2. Conditional
3. Independent event
4. chain rule
5. Bayes Theorem

✓ complete → প্রকৃষ্টতা বৈশিষ্ট্যের দ্বিত্ব
✓ consistent → কোনও এক নিয়ম assignment
→ নাও violate না করবে।

Binary esp : → variable
constraint
↓
৩ টি constrain two variable
বিলম্বিত।

□ Variety of constraints

1. unary
2. Binary
3. Higher order → constrain
involve 3 or more variable

Real world esp

1. Assignment problem
2. Timetable prob
3. Transportation schedule
4. Factory schedule

state are
defined by
the values assign

1. Initial state
2. Successor func
3. Goal test

constraint satisfaction problem এক - ০৬ video (BUET)

1. esp (what)
2. Backtracking search for esp
3. local search for esp.

State variable
Goal test Domain

classical problem → variable
esp prob → variable Domain

Map coloring

variable WA, VT, A, NSW, V, SA, T
Domain $D_i = \{red, green, blue\}$
constraints : neighbouring state not same
color.

সোমসং: প্রতিযোগিতা ২৫ জুন ২০১৭
কর (মহা) প্রদান

CH-08 Mahubub

এর universal কিনা আর existential সম্মান চলেছে
পরিচয় লাগলে আরো পরিচয় হতে পারে। কিন্তু universal
এর existential সম্মান (এর সম্মান-এর পরিচয় বসলে) এক
ভাষা পরিচয় হতে পারে।

~~github link~~ github link

unary relation
binary

✓ Sentence
✓ atomic sentence

→ Figure 8.3

FOPL
video — youtube

স্বাধীনতা
আন্দোলন

all boys are smart
if x is a boy then x is smart

→ $\forall x [B(x) \rightarrow S(x)]$

some girls are intelligent

x is a girl and x is intelligent

$\exists x [G(x) \wedge I(x)]$

⇒ there is a person who loves everyone
x love y

$\exists x \forall y \text{ love}(x, y)$

Masud sir

CR-18

18.5.5

Male / Female

N/H

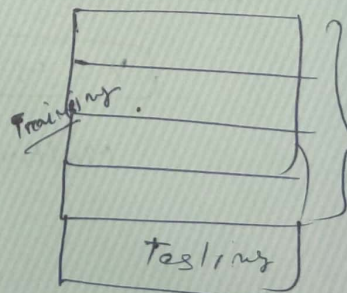
↓
class problem.

⇒ training

⇒ testing

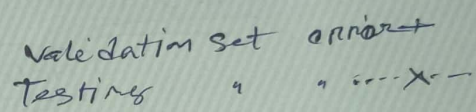
overfitting underfitting training

✓ K-fold cross validation



18.4.1 Model selection

Fig - 18.9



7 node base tree

18.6

18.6.1 univariate

$$y = w_1 x + w_0$$

1 मी द्वय द्वारा represent
करा पाय ।

Multivariate

→ ଆନୁଷ୍ଠାନିକ ସ୍ତର ଲିପି represent
କରିଥାଏ

minimize loss

↳ hill climbing
gradient descent algo

Multivalu

18.6.3 → Hard threshold hold

soft " / logistic regression

$$\text{logistic}(z) = \frac{1}{1 + e^{-z}}$$

Figure 18.17

NN (Single layer ANN
Multi " FFNN)

Non parametric model / Instan based

Nearest neigh basd model

K nearest neighbour

Support vector machine.

Fig - 18.30

ensemble learning

Probable of will climbing.



→ 1st 2nd chapter

W CH - 03 → [1 set]

in bar/uniform
n^o, uniform
d. iffer.
d. omitted

CH - 04 →

High elimb Prob.
Simu
loc
genetic algo

1st
CH - 06 →

Prob format

Map

Ciptagna exan

Andku exan

W CH - 13

→ 1 set condition

basic each
who to const v. basis
interpret -
common characteristics

W CH - 18

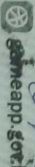
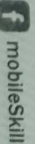
→ 1 set

Supportive
graduate.

Hand 1 set

loop
gamepp. 500
K. Meant to with a

Learning
detection tree
Model selection
10-fold cross
training testing
No guess reg. thm.
Learn in etc (Seq. log)



Microprocessor

কিছু ত্রিখণ্ড আছে

৩ অংশ

অন্যবৈশিষ্ট্যের থেকে কিছু বৈশিষ্ট্য অন্য
নেতা চেয়েছেন হার ।

০৮	1
০৮	2
০৮	3
০৮	4
০৮	5
০৮	6
০৮	7
০৮	8
০৮	9

Architecture

৩২০

[Exercise ৩২০ ৩২০]

স্বাধীনতা ১২ করে কমান

০৮ → 1
 ০৮ → 2
 ০৮ → 3
 ০৮ → 4
 ০৮ → 5
 ০৮ → 6
 ০৮ → 7
 ০৮ → 8
 ০৮ → 9

Arithmetic Entailment

০৮ → ০৮
 ০৮ → ০৮
 ০৮ → ১৮

০৮ 12 → 12.1
 12.2.1

০৮ → ১২ → ১২.১, ১২.
 ১১ → ১১.১

০৮ → ১২ → ১২.১,